Vitaran
Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

PHYSICS

Class – 6

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1. Physical Quantities and Measurements

Solutions

WORKSHEET – 1

1. **Correct Answer : is A) Standard measurement of any physical quantity.**

   Explanation: A unit is a fixed standard used to measure any physical quantity like length, mass, time, etc.

2. **Correct Answer : is C) Time**

   Explanation: Time is a basic (fundamental) quantity. Energy, force, and velocity are derived from fundamental quantities.

3. **Correct Answer : is A) P = NU**

   Explanation: Physical Quantity (P) = Numerical value (N) × Unit (U).
   Example: 5 m → 5 is number, m is unit.

4. **Correct Answer : is D) Speed**

   Explanation: Speed = Distance ÷ Time.
   Since it depends on other quantities, it is derived.

5. **Correct Answer : is A) Mass of a brick**

   Explanation: Mass is a fundamental physical quantity. Speed, pressure, and heat energy are derived quantities.

6. **Correct Answer : is B) Speed of a train**

   Explanation: Speed depends on distance and time, so it is derived.

7. **Correct Answer : is B) A standard or known measure**

   Explanation: Measurement means comparing an unknown quantity with a fixed standard unit.

8. **Correct Answer : is B) Numerical value, Unit**

   Explanation: 5 is the numerical value and cm is the unit.

9. **Correct Answer : is C) Fundamental physical quantities**

   Explanation: Derived quantities like force, speed, energy are expressed using fundamental quantities.

10. **Correct Answer : is B) Distance**

    Explanation: Physical quantities must be measurable. Distance can be measured. Happiness and beauty cannot be measured physically.

WORKSHEET -2

1. **Correct Answer: D**

   Explanation: Large distances between cities or places are measured in kilometres.
   Smaller units like cm or m are not suitable for long distances.

2. **Correct Answer: B**

   Explanation: 1 centimetre contains 10 millimetres.
   Millimetre is a smaller unit than centimetre.

3. **Correct Answer: A**

   Explanation: 1 metre equals 100 centimetres.
   So, 3 × 100 = 300 centimetres.

4. **Correct Answer: D) Dimensions of a football Ground**
   Explanation: football ground is very large, so it is usually measured in meters (m), not centimeters.
5. **Correct Answer: A**
   Explanation: The diameter is twice the radius.
   So, radius is half of the diameter.
6. **Correct Answer: A**
   Explanation: A credit card is very thin in thickness.
   Its thickness is approximately 1 millimetre.
7. **Correct answer: A) 5 cm**
   Explanation: 1 meter = 100 cm
   So,
   0.05 m = 0.05 × 100 = 5 cm
8. **Correct Answer: Cannot be determined**
   NA
9. **Correct Answer: B**
   Explanation: 1 metre equals 100 centimetres.
   So, 4.7 × 100 = 470 centimetres.
10. **Correct Answer: B**
    Explanation: 1 kilometre equals 1000 metres.
    So, 2.75 × 1000 = 2750 metres.

WORKSHEET-3

1. **Correct Answer: D) Simple pendulum**
   Explanation: A simple pendulum measures time by its regular oscillations. The other options measure length, not time.
2. **Correct Answer: B) 2400**
   Explanation: 1 minute = 60 seconds. So, 40 × 60 = 2400 seconds.
3. **Correct Answer: B) February**
   Explanation: In a leap year, February has 29 days instead of 28. Other months remain unchanged.
4. **Correct Answer: C) 1000**
   Explanation: 1 metric tonne equals 1000 kilograms. It is a standard SI unit conversion.
5. **Correct Answer: C) 1.5**
   Explanation: 90 seconds ÷ 60 = 1.5 minutes. There are 60 seconds in 1 minute.
6. **Correct Answer: B) 3000**
   Explanation: 1 second = 1000 milliseconds. Therefore, 3 × 1000 = 3000 ms.
7. **Correct Answer : B) 1000 g**
   Explanation: 1 quintal = 100 kg = 100,000 g. 1/100 of 100,000 g = 1000 g.
8. **Correct Answer: C) 1/100 kg**
   Explanation: 1 decagram = 10 g = 0.01 kg. 0.01 kg is equal to 1/100 kg.
9. **Correct Answer: B) 4000**
   Explanation: 1 millennium = 1000 years. So, 4 × 1000 = 4000 years.
10. **Correct Answer: C) 34 kg**
    Explanation: 1 hectogram = 100 g. 340 hg = 34,000 g = 34 kg.

WORKSHEET-4

1. **Correct Answer is A**
   100 m × 100 m = 10,000 m^2, which equals 1 hectare.
2. **Correct Answer is B)**
   1 m^2 = 1,000,000 mm^2
   50,000 mm^2 ÷ 1,000,000 = 0.05 m^2
3. **Correct Answer is B**
   Square millimetre is the smallest unit among mm^2, cm^2, m^2, and km^2.
4. **Correct Answer is D**
   1 hectare ≈ 2.47 acres.

5. **Correct Answer : is B**
Area = Length × Breadth
= 2.5 × 1.2 = 3.0 m^2

6. **Correct Answer : is C**
Hectare (10,000 m^2) is larger than acre (~4,047 m^2), square metre, and square centimetre.

7. **Correct Answer : is C**
The SI unit of area is square metre (m^2).

8. **Correct Answer : is B**
Volume means the space occupied by a substance (solid, liquid, or gas).

9. **Correct Answer : is D**
1 cm^3 means a cube with length, breadth, and height each 1 cm.

10. **Correct Answer : is C**
Volume of a rectangular box = Length × Breadth × Height.

WORKSHEET-5

1. **Correct answer:**
D. 25 kg

2. **Correct answer: B.**
When the unit is named after a person. Units like Newton (N), Pascal (Pa) use capital letters because they are named after scientists.

3. **Correct answer: D.**
By adding an 's' to the full name, like '5 kilograms'.
Full names can be plural (kilograms), but symbols never change (kg, not kgs).

4. **Correct answer: A. N·m**
A dot (·) is used to show multiplication of units clearly.

5. **Correct answer: B.**
Always lowercase, e.g., 'one newton of force.'
Full unit names are written in lowercase even if named after a person.

SINGLE CORRECT OPTION TYPE

1. **Correct Answer : is C) 1 km > 1 m > 1 cm > 1 mm**
Explanation: Kilometer is the largest unit and millimeter is the smallest. Units decrease by powers of 10.

2. **Correct Answer : is B) Area**
Explanation: Area is the number of unit squares that cover a surface. It measures surface extent.

3. **Correct Answer : is A) 500**
Explanation: 1 cm^2 = 100 mm^2. Therefore, 5 × 100 = 500 mm^2.

4. **Correct Answer : is D) 40**
Explanation: 1 kg = 10 hg. So, 4 × 10 = 40 hg.

5. **Correct Answer : is A) 1/15**
Explanation: 1 hour = 3600 seconds. 240 ÷ 3600 = 1/15 hour.

6. **Correct Answer : is B) 3000**
Explanation: 1 minute = 60 seconds. 50 × 60 = 3000 seconds.

7. **Correct Answer : is A) Sundial**
Explanation: A sundial works using sunlight and shadows. It cannot work at night or without Sun.

8. **Correct Answer : is A) 5.7 cm**
Explanation: Length = 6.9 − 1.2 = 5.7 cm. Subtraction gives the correct measurement.

9. **Correct Answer : is C) 40 m**
Explanation: Unit symbols are not plural and are written in lower case if not named after a scientist.

10. **Correct Answer : is B) Sidereal day**
Explanation: A sidereal day is Earth's rotation with respect to fixed stars. It is slightly shorter than a solar day.

11. Correct Answer : is C) Length — Fundamental quantity

Explanation: Length is a basic physical quantity. Speed, area, and density are derived quantities.

12. Correct Answer : is D) Atomic clock

Explanation: Atomic clocks use atomic vibrations for precise measurement. They are the most accurate time devices.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

13. Correct Answer : is B, C

Explanation: Fundamental quantities are independent. They do not depend on other physical quantities.

14. Correct Answer : is A, B

Explanation: Hand span and cubit vary from person to person. They are not standard units.

15. Correct Answer : is B, C

Explanation: 1 pound ≈ 0.456 kg = 456 g.

16. Correct Answer : is A, B

Explanation: 1 hour = 60 minutes = 3600 seconds.

17. Correct Answer : is C) 1 cm^2

Explanation: 5 cm = 50 mm. Area = 2 mm × 50 mm = 100 mm^2 = 1 cm^2.

18. Correct Answer : is A, B

Explanation: K = °C + 273 ⟶ 35 + 273 = 308 K. Also equals 95°F.

19. Correct Answer : is A, B, C

Explanation: Kelvin, square metre, and kilogram are SI units. Gram is not SI base unit.

ASSERTION AND REASON TYPE

20. Correct Answer : is A

Explanation: Area is derived from length × length. So m^2 is correct SI unit.

21. Correct Answer : is A

Explanation: Mass is fundamental and measured directly. It does not depend on other quantities.

22. Correct Answer : is C

Explanation: 1 km = 1000 m is true. But kilometre is used for long distances, not short classroom widths.

23. Correct Answer : is A

Explanation: Liquids are commonly measured in mL and solids in cm^3. Both statements are correct and related.

MATRIX MATCHING TYPE

24. Correct Answer : is:

A – S, B – T, C – Q, D – R

Explanation:
1 hour = 3600 s
1 day = 86400 s
1 decade = 10 years
1 century = 100 years

25. Correct Answer : is:

A – S, B – P, C – Q, D – R

Explanation:

Force ⟶ Newton (N)
Mass ⟶ kg
Power ⟶ Watt (W)
Length ⟶ metre (m)

COMPREHENSION TYPE

26. Correct Answer : is A) 10

Explanation: Magnitude means numerical size of the quantity. Here it is 10.

27. Correct Answer : is A) 10

Explanation: Numerical value is the number without unit. It is 10.

28. Correct Answer : is B) cm^2

Explanation: Unit part of 10 cm^2 is cm^2.

29. **Correct Answer : is C) 5,00,000**

Explanation: 5 km = 5000 m = 5000 × 100 = 5,00,000 cm.

30. **Correct Answer : is B) 6,000**

Explanation: 600 × 10 = 6000 mm.

31. **Correct Answer : is A) 200**

Explanation: 1 cm^2 = 100 mm^2. 20000 ÷ 100 = 200 cm^2.

INTEGER TYPE

32. **Correct Answer : is 5**

Explanation: 1/2000 km = 0.5 m = 50 cm = 5 × 10 cm.

33. **Correct Answer : 1 gram = 1000 milligrams.**

34. **Correct Answer : is 3600**

PREVIOUS YEAR QUESTIONS

1. **Correct Answer : *Answer is D*) 14.2 cm $-$ 5 cm $= 9.2$ cm**

Final Answer: 9.2 cm

2. **Correct Answer : (Option D)**

- Starting point = 2.0 cm
- Ending point = 14.2 cm

So, total length of thread =

$$14.2 - 2.0 = 12.2 \text{ cm}$$

It makes 6 rounds, so length for 1 round =

$$\frac{12.2}{6} = 2.03 \text{ cm}$$

Final Answer: 2.03 cm

3. **Correct Answer : is A)**

Trisha's distance = 500 m

Ansh's distance = 0.6 km = 600 m

- Both walk at the same speed (same footsteps), so the one with less distance reaches first.
- Trisha reaches first.

4. **Correct Answer : The correct option is C.**

- Apparent length: 38 mm
- Actual length: 30 mm

5. **Correct Answer : 1 metre = 100 centimetres**

1 centimetre = 10 millimetres

So the blanks are:

- 100
- centimetre
- 10
- millimetre

6. **Correct Answer: D. I and IV only**

I. 40 cm = 4 dm

True (1 dm = 10 cm → 4 dm = 40 cm)

IV. 10 m = 100 dm

True (1 m = 10 dm → 10 m = 100 dm)

7. **Correct Answer : Total length of wire = 18 cm = 180 mm**

Length used in one turn = 0.04 mm

Number of turns = Total length ÷ Length per turn

$$\text{Turns} = \frac{180}{0.04} = 4500$$

2. Temperature and its Measurement

Solutions

WORKSHEET – 1

1. **Correct Answer : B) Scalar quantity**
   Explanation: Temperature tells only how hot or cold a body is. It has no direction, so it is a scalar quantity.
2. **Correct Answer : B) Temperature increases**
   Explanation: When heat is given to a body, its temperature usually increases (if it does not change state).
3. **Correct Answer : C) Kelvin**
   Explanation: The SI unit of temperature is Kelvin (K). For Class 6, degree Celsius is commonly used in daily life, but SI unit is Kelvin.
4. **Correct Answer : B) From higher temperature to lower temperature**
   Explanation: Heat always flows from a hotter body to a colder body.
5. **Correct Answer : B) The amount of heat energy supplied or removed**
   Explanation: Temperature changes when heat is added or removed from a body.
6. **Correct Answer : C) Heat is cause and temperature is effect**
   Explanation: When heat is given (cause), temperature changes (effect).
7. **Correct Answer : C) Joule**
   Explanation: Heat is a form of energy, and the SI unit of energy is joule (J).
8. **Correct Answer : C) It does not state the exact degree of coldness**
   Explanation: In science, temperature should be given with a number and unit (like 5°C). Saying "cold" is not exact.
9. **Correct Answer : C) Low**
   Explanation: A cold body has low temperature.
10. **Correct Answer : A) Temperature determines direction of heat flow**
    Explanation: Heat flows from a body at higher temperature to a body at lower temperature.

WORKSHEET-2

1. **Correct Answer : B) Clinical thermometer**
   Explanation: A clinical thermometer is specially designed to measure human body temperature (about 35°C to 42°C).
2. **Correct Answer : C) –10 °C to 110 °C**
   Explanation: A laboratory thermometer measures a wide range of temperatures, usually from –10°C to 110°C.
3. **Correct Answer : A) Clinical**
   Explanation: A kink is present in a clinical thermometer to prevent mercury from falling back quickly, so the reading can be noted.
4. **Correct Answer : 4: D) All the above**
   Explanation: Temperature can be measured using Celsius (°C), Fahrenheit (°F), and Kelvin (K) scales.

5. **Correct Answer : A) 0 °C**

Explanation: Conversion formula:
°C = K − 273
= 273 − 273 = 0°C

6. **Correct Answer : C) Kelvin Scale**

Explanation: Kelvin scale measures absolute temperature. 0 K is absolute zero.

7. **Correct Answer : B) 300 K**

Explanation: K = °C + 273
= 27 + 273 = 300 K

8. **Correct Answer : B) 101.5 °F**

Explanation: Normal body temperature is about 98.6°F. A temperature above 100°F indicates fever.

9. **Correct Answer : D) 59**

Explanation: °F = (9/5 × °C) + 32
= (9/5 × 15) + 32
= 27 + 32 = 59°F

10. **Correct Answer : A) 65 °C**

Explanation: °C = K − 273
= 338 − 273 = 65°C

11. **Correct Answer : (B) To show approximate room air temperature**

Explanation: Thermometers placed on walls measure the surrounding air temperature of the room, giving an approximate value of the indoor environment.

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE1:

1. **Correct Answer : (C) Skin senses flow of heat, not temperature directly**

Explanation: Our skin acts as a detector of heat transfer (heat flow), not absolute temperature. A metal object feels colder than a wooden one at the same room temperature because metal conducts heat away from the hand faster.

2. **Correct Answer : C) Its temperature is higher than our body**

Explanation: Heat flows from a region of higher temperature to lower temperature. If an object feels hot, it means heat is flowing from the object into our body, indicating $T_{\text{object}} > T_{\text{body}}$.

3. **Correct Answer : C) Laboratory thermometer**

Explanation: Boiling water is at $100°C$. A clinical thermometer only ranges from approximately $35°C$ to $42°C$ and would break. A laboratory thermometer ($-10°C$ to $110°C$) covers the required range.

4. **Correct Answer : B) Expansion and contraction of substances due to heating and cooling**

Explanation: Mercury expands (volume increases) when heated and contracts (volume decreases) when cooled. This change in volume drives the liquid up or down the capillary tube.

5. **Correct Answer : C) It has no kink**

Explanation: A laboratory thermometer lacks a "kink" or constriction. Without it, the mercury level drops immediately due to gravity and cooling as soon as it is removed from the source, making it impossible to read accurately after removal.

6. **Correct Answer : B) Laboratory thermometer**

Explanation: The standard range for a laboratory thermometer used in schools is typically $-10°C$ to $110°C$, allowing measurement of freezing points, boiling points, and chemical reactions.

7. **Correct Answer : C) Measure temperature without physical contact**
Explanation: Infrared (IR) thermometers detect thermal radiation emitted by the body. This allows non-contact screening, preventing the spread of infection during a pandemic.

8. **Correct Answer : C) Fahrenheit**
Explanation: The Fahrenheit scale is defined by the freezing point of water at $32°F$ and the boiling point at $212°F$.
Difference: $212 - 32 = 180$ divisions
(Celsius and Kelvin have 100 divisions.)

9. **Correct Answer : B) It gives incorrect reading**
Explanation: If the bulb touches the bottom or sides of the container, it measures the temperature of the container rather than the actual temperature of the liquid.

10. **Correct Answer : C) Mercury falls back immediately**
Explanation: Since a laboratory thermometer has no kink, the mercury contracts instantly when removed from the hot liquid into cooler air, leading to an error in reading.

11. **Correct Answer : C) Lowest temperature during the day**
Explanation: In weather reports, the minimum temperature refers to the lowest temperature recorded within the 24-hour cycle.

12. **Correct Answer : B) 25°C**
Explanation: Formula: $C = (F - 32) \times \frac{5}{9}$
Calculation: $77 - 32 = 45$
$$45 \times \frac{5}{9} = 25°C$$

13. **Correct Answer : C) 100°C**
Explanation: $212°F$ corresponds to $100°C$ (boiling point of water at sea level).

14. **Correct Answer : A) 0 K**
Explanation: Formula: $K = C + 273$
Calculation: $-273 + 273 = 0\ K$
(Absolute zero)

15. **Correct Answer : A) 5°C**
Explanation: Formula: $C = (F - 32) \times \frac{5}{9}$
Calculation: $41 - 32 = 9$
$$9 \times \frac{5}{9} = 5°C$$

16. **Correct Answer : B) 10 K**
Explanation: $\Delta C = \Delta K$
Change in Celsius: $35 - 25 = 10°C$
Therefore, increase = $10\ K$

17. **Correct Answer : B) Both bodies are at the same temperature**
Explanation: If no heat flows, the bodies are in thermal equilibrium:
$$T_1 = T_2$$

18. **Correct Answer : Kelvin and Celsius have same interval size**
Explanation: $K = C + 273$
For every increase of 1°C, Kelvin also increases by 1 K. The scales differ only in their zero points.

19. **Correct Answer : B) Degree of hotness or coldness of a body**
Explanation: This is the standard textbook definition of temperature.

20. **Correct Answer : B) Inform about climatic conditions**
Explanation: Daily maximum and minimum temperatures help indicate weather and climatic conditions.

21. **Correct Answer : (C)**
The clinical thermometer range ($\mathbf{35°C}$ to $\mathbf{42°C}$)is designed specifically for human body measurement.

ONE OR MORE THAN CORRECT OPTION TYPE

22. **Correct Answer : A, B, C, D**
Explanation:
A) $\frac{C}{5} = \frac{F-32}{9}$ is correct relation.
B) $K = C + 273$.
C) 0 K = −273°C.
D) Kelvin temperature is called absolute temperature.

23. Correct Answer : A, B

Explanation:

25°C →

°F = $\frac{9}{5} \times 25 + 32 = 77°F$

K = 25 + 273 = 298 K

24. Correct Answer : A, C

Explanation:

Celsius: 0°C to 100°C → 100 divisions.

Kelvin: 273 K to 373 K → 100 divisions.

Fahrenheit = 180 divisions; Reaumur = 80 divisions.

25. Correct Answer : A, B, C

Explanation:

Fahrenheit: 180 divisions (largest).

Reaumur: 80 divisions (smallest).

Celsius & Kelvin: 100 divisions (equal).

Fixed points differ in numbers, so D is incorrect.

26. Correct Answer : B, C

Explanation:

Upper fixed point:

Celsius = 100

Fahrenheit = 212

Kelvin = 373

Reaumur = 80

Only Fahrenheit and Kelvin exceed 200 units.

ASSERTION AND REASON TYPE

27. Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Heat flows from higher temperature to lower temperature; hence temperature difference determines direction of heat flow.

28. Correct Answer : B) Both A and R are true but R is not the correct explanation of A

Explanation: Temperature is scalar because it has magnitude only (no direction). Being measured in °C, K, °F does not explain why it is scalar.

29. Correct Answer : D) A is false but R is true

Explanation: Laboratory thermometer is not used for body temperature. It has no kink; hence mercury falls back immediately.

30. Correct Answer : C) A is true but R is false

Explanation: Digital thermometers are safer as they do not contain mercury. However, they do contain batteries; so Reason is false.

MATRIX MATCHING TYPE

31. Correct Answer : a) 32°F → R) Lower fixed point of Fahrenheit scale

Explanation: 32°F is the freezing point of water in the Fahrenheit scale, called the lower fixed point.

b) −273°C → P) Absolute zero temperature

Explanation: −273°C is the lowest possible temperature where all molecular motion stops (absolute zero).

c) 100°C → S) Equal to 373 K

Explanation: 100°C is the boiling point of water and equals 373 Kelvin (K = °C + 273).

d) 0°C → Q) Equal to 273 K

Explanation: 0°C is the freezing point of water and equals 273 Kelvin.

32. Correct Answer : a) Mercury expansion → Q) Based on heating and cooling effect

Explanation: Mercury expands when heated and contracts when cooled, which helps measure temperature.

b) Clinical thermometer scale → R) 35°C to 42°C range

Explanation: Clinical thermometers are designed to measure human body temperature, usually between 35°C and 42°C.

c) Laboratory thermometer position → S) Must be kept vertical while measuring

Explanation: A laboratory thermometer should be held upright to get an accurate reading.

d) Infrared thermometer → P) Measures temperature without contact

Explanation: Infrared thermometers detect heat radiation, so they can measure temperature without touching the object.

COMPREHENSION TYPE

33. Correct Answer : A) Celsius and Kelvin scales have equal number of divisions

Explanation:

Celsius: 100 − 0 = 100 divisions

Kelvin: 373 − 273 = 100 divisions

Hence both have equal divisions.

34. Correct Answer : B) Because 212 − 32 = 180

Explanation:

Number of divisions = Upper fixed point − Lower fixed point

= 212 − 32 = 180

35. Correct Answer : A) It equals 212°F and 373 K

Explanation:

°F = $\frac{9}{5} \times 100 + 32 = 212°F$

K = 100 + 273 = 373 K

36. Correct Answer : C) It avoids physical contact

Explanation: Infrared thermometer measures temperature without touching the body, reducing infection spread.

37. Correct Answer : C) Mercury level falls immediately

Explanation: Laboratory thermometer has no kink; mercury falls back when removed, causing wrong reading.

38. Correct Answer : C) Kink (constriction)

Explanation: The kink prevents mercury from flowing back immediately, so reading remains fixed.

INTEGER TYPE

39. Correct Answer : 35°C

Explanation: $°C = \frac{5}{9}(°F - 32)$

= $\frac{5}{9}(95 - 32)$

= $\frac{5}{9}(63) = 35°C$

40. Correct Answer : 383 K

Explanation:$K = °C + 273$

= 110 + 273 = 383 K

41. Correct Answer : 80 divisions

Explanation: Upper fixed point = 80°R

Lower fixed point = 0°R

Total divisions = 80 − 0 = 80

42. Correct Answer : 27°C

Explanation: $°C = K - 273$

= 300 − 273 = 27°C

PREVIOUS YEAR QUESTIONS

1. Correct Answer : C) 0°C

The thermometer shows −2°C at 0°C and 98°C at 100°C.

So, total span = 98 − (−2) = 100°C (correct scale).

Only a constant error (offset) is present, which cancels when finding temperature difference.

Therefore, no error in temperature difference = 0°C.

2. Correct Answer : C) 1.0°C lower than the true temperature

The thermometer reads −1°C instead of 0°C → −1°C error (zero error).

Span is still correct (99 − (−1) = 100°C), so only a constant shift exists.

Every reading will be 1°C less than actual.

3. Motion

Solutions

WORKSHEET – 1

1. **Correct Answer : (B) When its position changes with time**

   Explanation: Motion is defined as a change in the position of an object over time relative to a reference point.

2. **Correct Answer : (C) A book kept on a desk**

   Explanation: An object is at rest if its position does not change relative to its surroundings. The book remains stationary on the desk.

3. **Correct Answer : (C) Rectilinear motion**

   Explanation: Motion of an object along a straight line is called rectilinear or linear motion.

4. **Correct Answer : (A) Rotatory motion**

   Explanation: The blades move around a fixed axis, which characterizes rotatory motion.

5. **Correct Answer : (B) Oscillatory motion**

   Explanation: The to-and-fro motion of an object about its mean position is called oscillatory motion.

6. **Correct Answer : (B) Periodic motion**

   Explanation: Any motion that repeats itself at regular intervals of time T is termed periodic motion.

7. **Correct Answer : (A) Circular motion**

   Explanation: The stone follows a circular path around a central point (the hand) at a constant distance.

8. **Correct Answer : (C) Random motion**

   Explanation: Fish change direction and speed irregularly, which is characteristic of random or non-linear motion.

9. **Correct Answer : (C) Random motion**

   Explanation: Motion without a specific path or constant direction in a crowded space is defined as random motion.

10. **Correct Answer : (B) To compare position**

    Explanation: Motion is judged by comparing position with a reference point.

WORKSHEET-2

1. **Correct Answer : (C) Distance**
   Explanation:

   The total length of the path a person actually walks, no matter the direction, is simply called distance. Formula: Distance = Total path covered

2. **Correct Answer : (A) 10 m/s east**
   Explanation:

   He travels a total of 200 meters east in 20 seconds total. You divide the total displacement by the total time. Formula: Average Velocity = Total Displacement / Total Time = 200 m / 20 s = 10 m/s east

3. **Correct Answer : (C) Displacement**
   Explanation:

   While distance is the actual path taken, displacement is just a straight line drawn directly from the start to the finish line. Formula: Displacement = Final position - Initial position

4. **Correct Answer : (C) 0 m Explanation:**

   Because the boy walks out and comes right back to the exact spot he started from, his overall change in position is nothing. Formula: Displacement = 0 (when start and end points are the same)

5. **Correct Answer : (A) 600 m Explanation:**

To find how far the bus went, you multiply how fast it was going by how long it traveled. Formula: Distance = Speed x Time = 20 m/s x 30 s = 600 m

6. **Correct Answer : (C) time Explanation:**

Speed tells us how fast an object moves, which we figure out by seeing how much distance it covers in a certain amount of time. Formula: Speed = Distance / Time

7. **Correct Answer : (A) 10 m/s Explanation:**

We find the speed by dividing the total distance the car moved by the time it took to move that far. Formula: Speed = Distance / Time = 90 m / 9 s = 10 m/s

8. **Correct Answer : (B) Speedometer Explanation:**

A speedometer is the dial on a car or bike's dashboard that tells you exactly how fast you are moving at that very second.

9. **Correct Answer : (A) total time Explanation:**

To find the average speed for a whole trip, you take the complete distance traveled and divide it by the complete time it took. Formula: Average Speed = Total Distance / Total Time

10. **Correct Answer : (A) 5 m/s north Explanation:**

Velocity is just speed with a specific direction attached to it, found by dividing displacement by time. Formula: Velocity = Displacement / Time = 150 m / 30 s = 5 m/s North

11. **Correct Answer : (B) 20 m/s**

Explanation: For uniform acceleration, average velocity = $\frac{u+v}{2}$.

= $\frac{10+30}{2} = 20$m/s

12. **Correct Answer : (C) 13 m/s**

Explanation: Average velocity = $\frac{u+v}{2}$.

= $\frac{8+18}{2} = 13$m/s

COMPETITIVE CORNER

SINGLE CORRECT OPTION TYPE1:

1. **Correct Answer : (B) Reference point**

Explanation: Motion is a relative concept; an object is judged to be in motion or at rest by comparing its position to a stationary background or reference point.

2. **Correct Answer : (C) Other passengers in the bus**

Explanation: Relative to the bus and the other passengers moving at the same velocity, the person's position does not change over time.

3. **Correct Answer : (A) Translatory motion**

Explanation: The entire body of the balloon moves from one point in space to another without necessarily rotating, which fits the definition of translatory motion.

4. **Correct Answer : (B) Curvilinear motion**

Explanation: A thrown ball follows a curved projectile arc; since the path is a curve, the motion is curvilinear.

5. **Correct Answer : (B) Rotatory motion**

Explanation: The hands of the clock turn continuously around a central, internal pivot point (a fixed axis).

6. **Correct Answer : (C) Earth moving around the Sun**

Explanation: Earth completes an orbit in a fixed time interval (periodic) but does not swing back and forth across a mean equilibrium point (oscillatory).

7. **Correct Answer : (B) Vibratory motion**

Explanation: Striking the membrane causes rapid, localized back-and-forth movements (vibrations) that produce sound.

8. **Correct Answer : (B) Periodic motion**

Explanation: By definition, periodic motion is any motion that repeats itself in equal, regular intervals of time.

9. Correct Answer : (C) Circular motion

Explanation: While the wheel rotates, each seat travels in a circular path around the central axis, showing circular motion.

10. Correct Answer : (B) Fixed axis

Explanation: Rotatory motion requires an object to spin around a stationary line or pivot point called a fixed axis.

11. Correct Answer : (C) 20 s

Explanation: Using the formula $t = \frac{d}{s}$, we calculate $t = \frac{300}{15} = 20$s.

12. Correct Answer : (C) 100 m

Explanation: Using the formula $d = s \times t$, we calculate $d = 2 \times 50 = 100$m.

13. Correct Answer : (C) 60 km/h

Explanation: Using the formula for average speed $v = \frac{\text{Total Distance}}{\text{Total Time}}$, we calculate $v = \frac{120}{2} = 60$km/h.

14. Correct Answer : (C) 200 km

Explanation: Using the formula $d = s \times t$, we calculate $d = 50 \times 4 = 200$km.

15. Correct Answer : (B) 2 m/s

Explanation: Using the formula $s = \frac{d}{t}$, we calculate $s = \frac{400}{200} = 2$m/s.

16. Correct Answer : (C) 150 m

Explanation: For straight-line motion without direction change, displacement equals distance. Using Displacement $= v \times t$, we calculate $5 \times 30 = 150$m.

17. Correct Answer : (A) 6 m/s north

Explanation: Using the formula for velocity $v = \frac{\text{Displacement}}{t}$, we calculate $v = \frac{90}{15} = 6$m/s north.

18. Correct Answer : (B) 10 m/s east

Explanation: Total displacement = 100 + 100 = 200 m east. Total time = 10 + 10 = 20 s. Using $v = \frac{\text{Total Displacement}}{\text{Total Time}}$, we calculate $v = \frac{200}{20} = 10$m/s east.

ONE OR MORE THAN CORRECT OPTION TYPE

19. (Correct Answer :B) Stone falling vertically;

(D) elevators moving vertically in a shaft
Explanation: Rectilinear motion is motion along a straight line; vertical fall and vertical lift motion follow a straight path.
Validity: Valid

20. Correct Answer : (A) Second hand of a clock;

(B) Blades of a ceiling fan; (D) a merry-go-round
Explanation: Circular motion means moving along a circular path around a fixed center.
Validity: Valid

21. Correct Answer : (A), (B), (D)

Explanation: Distance = total path length travelled; it depends on path and SI unit is metre; it does not include direction.
Validity: Valid

22. Correct Answer : (A), (B), (C)

Explanation: Velocity $= \frac{\text{displacement}}{\text{time}}$; it has magnitude and direction and is a vector quantity.
Validity: Valid

23. Correct Answer : (A), (B)

Explanation: Speed $= \frac{\text{distance}}{\text{time}}$; it has no direction and is scalar; it cannot be negative at Class-6 level.
Validity: Valid

ASSERTION AND REASON TYPE

24. Correct Answer : (A) Both A and R are true and R correctly explains A

Explanation: An object is at rest when its position does not change with time relative to surroundings — this is the definition of rest.

25. Correct Answer : (C) A is true but R is false

Explanation: Rectilinear motion means motion along a straight line, not a curved path; the assertion is correct but the reason is wrong.

26. Correct Answer : (D) A is false but R is true

Explanation: A pendulum shows oscillatory motion (to and fro about mean position), not circular motion; the reason correctly defines oscillatory motion.

27. Correct Answer : (D) A is false but R is true

Explanation: Distance is total path length and is always positive (≥ 0); it cannot be negative.

MATRIX MATCHING TYPE

28. Correct Answer : a–R, b–P, c–Q, d–S

Explanation: Wheel spinning + forward motion → combined rotatory & translatory; mosquito changes path → random; guitar string vibrates → vibratory; Earth revolution repeats in fixed time → periodic.

29. Correct Answer :a–S, b–P, c–R, d–Q

Explanation: Distance is total path length and never negative; displacement can be zero after motion; speed is measured by speedometer; velocity has direction.

COMPREHENSION TYPE

30. Correct Answer : (A) Periodic motion

Explanation: The spring toy repeats up–down motion at equal intervals — periodic motion.

31. Correct Answer : (C) Combined rotatory and translatory motion

Explanation: Wheels rotate while the car's body moves forward — rotation + translation together.

32. Correct Answer : (B) Rotatory motion

Explanation: The windmill rotates about a fixed rod (axis).

33. Correct Answer : (C) 30 m

Explanation: Total distance = 20 m + 10 m = 30 m.

34. Correct Answer : (C) Zero

Explanation: Displacement = final position – initial position; if he returns home, net change = 0.

35. Correct Answer : (B) Direction

Explanation: Velocity = speed with direction.

INTEGER TYPE

36. Correct Answer :11

Explanation: Total displacement = 8 m east + 3 m east = 11 m (same direction).

37. Correct Answer :0

Explanation: After one full round, final position = initial position ⇒ displacement = 0 m.

38. Correct Answer :6

Explanation: Using $v = \frac{d}{t}$, $v = \frac{90}{15} =$ 6m/s.

39. Correct Answer :25

Explanation: Speed $= \frac{90}{15} = 6$m/s; time $t = \frac{d}{v} = \frac{150}{6} = 25$s.

PREVIOUS YEAR QUESTIONS

1. Correct Answer : (c)

(a) – (ii) Periodic motion
The Earth revolving around the Sun repeats its motion after a fixed time (1 year), so it is periodic motion.

(b) – (ii) Periodic motion
The clock hands repeat their positions after equal intervals of time, so this is also periodic motion.

(c) – (iii) Oscillatory motion
The pendulum moves to and fro about its mean position, so it shows oscillatory motion.

Thus, the correct option is C.

2. Correct Answer : (B)

I. The motion of the rider along with the bicycle is rectilinear motion. True (T). When a bicycle moves along a straight path, the entire system (rider and bike) moves forward in a straight line.

II. The motion of the feet pedaling is always periodic. False (F). While pedaling *can* be periodic if you maintain a perfectly constant rhythm, it is not "always" periodic. If the rider speeds up, slows down, or coasts without pedaling, the motion loses its fixed time interval.

III. The wheel of a moving bicycle shows circular motion only. False (F). This is a common trick! A moving wheel actually exhibits combination motion. It has rotatory motion (spinning around the axle) AND rectilinear motion (moving forward along the road).

IV. The motion of a potter's wheel is rotatory only. True (T). Unlike the bicycle wheel, the potter's wheel is fixed in one spot. It spins around a central axis but does not move from place to place.

3. Correct Answer : (A) Distance is the total path covered. If you move, it is always positive (finite).

Displacement is the shortest straight line between your start and end points.

Statement 1 (True): You can travel a long way (finite distance) but end up with zero displacement.

Statement 2 (True & Explains): It gives the reason for Statement 1. If you start and end at the same point (a round trip), your change in position is zero, so displacement is zero.

4. Correct Answer :

5. Correct Answer :Who will reach first?

Analysis:
Trisha's Distance = 500 m
Ansh's Distance = 0.6 km

First, convert Ansh's distance into metres to match the unit:
0.6 km × 1000 = 600 m

Comparison:
Both children take the same size footsteps (80 cm) and move at the same rate (same speed). Therefore, the child who has the shorter distance to travel will reach first.

500 m < 600 m

Conclusion:
Trisha has a shorter distance to cover, so she will reach the playground first.

Correct Option: A. Trisha

6. Correct Answer : B: I, II and III only

You need to find the incorrect statements:

I (Incorrect): The bag moves with the bus. Relative to the floor, it is at rest, not in motion.

II (Incorrect): Never use a worn-out zero mark. You should start from a clear mark (like 1 cm) to avoid errors.

III (Incorrect): Blinking is not oscillatory motion. Oscillatory motion is a repetitive to-and-fro swinging (like a pendulum).

IV (Correct): Moving around a central point is indeed called revolution.

4. Light, Shadows and Reflection

Solutions

WORKSHEET – 1

1. **Correct Answer : C) Energy**

   Explanation: Light is a form of electromagnetic radiation that carries energy. Energy relation: **E = hf**.

2. **Correct Answer : A) Luminous**

   Explanation: Stars emit their own light, so they are luminous sources. Luminous body → self-light emission.

3. **Correct Answer :A) 3,00,000 km/s**

   Explanation: Speed of light in vacuum is c = 3×10^8 m/s = 3×10^5 km/s.

4. **Correct Answer :A) Straight lines**

   Explanation: Light travels in straight lines in a uniform medium.

5. **Correct Answer :C) Glass**

   Explanation: Glass allows light to pass through, so it is an optical medium. Transmission property based.

6. **Correct Answer : D) Optical medium**

   Explanation: Any material allowing light wholly or partially is an optical medium. Includes transparent & translucent.

7. **Correct Answer : C) Light from the torch reflects from the box into our eyes**

   Explanation: Non-luminous objects are seen due to **reflection of light** into the eye

8. **Correct Answer : C) Divergent beam of light**

   Explanation: Rays spreading from a single point form a divergent beam. Source → outward spread.

9. **Correct Answer : C) Translucent medium**

   Explanation: Translucent materials allow **partial transmission + scattering** of light.

10. **Correct Answer : B) Transparent, Translucent, Opaque**

    Explanation: Clear glass → transparent; butter paper → translucent; wood → opaque based on transmission.

WORKSHEET-2

1. **Correct Answer : D) Shape of object**

   Explanation: A shadow mainly shows the outline/shape of an object, not color or details. Light blocking → silhouette formed.

2. **Correct Answer : D) Butter Paper**

   Explanation: Butter paper is translucent and produces blurred shadows due to partial transmission and scattering.

3. **Correct Answer : A) 2 cm**

   Explanation: Magnification = image height/object height = screen distance/object distance = 10/100. Image height = 20 × (10/100) = 2 cm. Formula: $h_i/h_o = v/u$

4. **Correct Answer : A) 8 min. 20 sec**

   Explanation: Sun–Earth distance ≈ 1.5×10^8 km; time = distance/speed = $(1.5\times10^8)/(3\times10^5) \approx 500$ s = 8 min 20 s. Formula: t = d/c

5. **Correct Answer : A) Umbra**

   Explanation: Umbra is the completely dark central region of a shadow where light is fully blocked.

6. **Correct Answer : B) lunar**

   Explanation: Lunar eclipse occurs when Earth comes between Sun and Moon in one straight line.
   Alignment: Sun–Earth–Moon.

7. **Correct Answer : B) noon**

Explanation: At noon the Sun is nearly overhead, so shadow length is minimum. Shadow length ∝ 1/tan(altitude).

8. **Correct Answer : B) Becomes larger**

Explanation: Bringing object closer to the light source increases shadow size due to larger angular spread.
Concept: Diverging rays → magnified shadow.

9. **Correct Answer : C) A sees total, B sees partial**

Explanation: Umbra region gives total eclipse; penumbra region gives partial eclipse.

10. **Correct Answer : C) Image becomes blurred**

Explanation: Large pinhole allows many rays → overlapping → loss of sharpness. Small hole → sharp image.

11. **Correct Answer : B) 4 cm**

Explanation: Image height = 16 × (20/80) = 4 cm using pinhole camera ratio.
Formula: $h_i/h_o = v/u$

WORKSHEET-3

1. **Correct Answer : C) Point of incidence**

Explanation: The exact point on the mirror where the incident ray strikes is called the point of incidence. It is the reference point for measuring angles.

2. **Correct Answer : B) Same plane**

Explanation: By the Second Law of Reflection, the incident ray, reflected ray, and the normal always lie in the same plane.

3. **Correct Answer : (A)**

4. **Correct Answer : D) Glancing angle of reflection**

Explanation: The angle between the reflected ray and mirror surface is glancing angle of reflection. Relation: θg = 90° − r.

5. **Correct Answer : B) Angle of incidence**

Explanation: The angle between the incident ray and the normal is called the angle of incidence (i).

6. **Correct Answer : A)35°**

Explanation: First Law of Reflection → angle of incidence equals angle of reflection. So r = i = 35°.

7. **Correct Answer : B) 65°**

Explanation: Glancing angle = 25° ⇒ i = 90° − 25° = 65°. Since i = r, reflection angle = 65°.

8. **Correct Answer : B) 55°**

Explanation: Law of reflection states i = r. If reflection angle is 55°, incidence angle is also 55°.

9. **Correct Answer : D) laterally inverted**

Explanation: A plane mirror image is virtual, erect, same size, and laterally inverted (left-right reversed).

10. **Correct Answer : A) 1**

Explanation: Plane mirror forms same-size image. Magnification m = hi/ho = 1.

11. **Correct Answer : C) 45°**

Explanation: In a periscope, mirrors are placed at 45° to the tube so light turns by 90° after reflection.

12. **Correct Answer : C) Multiple reflections**

Explanation: Kaleidoscope works by repeated reflections between inclined mirrors to form patterns.

13. **Correct Answer : B) 0°**

Explanation: In a standard periscope, emergent ray is parallel to incident ray, so angle between them is 0°.

14. **Correct Answer : A) 90°**

Explanation: At one 45° mirror, deviation = i + r = 45° + 45° = 90°.

SINGLE CORRECT OPTION TYPE

1. **Correct Answer : A — Polished surface**

Explanation: A mirror works by regular reflection from a smooth surface. Polished surfaces produce specular reflection where angle of incidence = angle of reflection (i = r).

2. Correct Answer : A — Translucent medium

Explanation: Oil (oiled) paper allows partial transmission of light. It follows partial transmission T < 1 and partial scattering.

3. Correct Answer : B — Extended source of light

Explanation: A source larger than a point is called an extended source. Its size ≫ wavelength and not treated as a single-point emitter.

4. Correct Answer : B — Glancing angle of reflection

Explanation: Angle with mirror surface = 90° − r. Since angle of reflection r is measured from normal, glancing angle = 90° − r.

5. Correct Answer : D

6. Correct Answer : C — Sun

Explanation: Earth's primary natural light source is the Sun. Solar radiation provides maximum light energy reaching Earth.

7. Correct Answer : B — Transparent medium

Explanation: Transparent media transmit most light energy. Transmission coefficient ≈ 1 compared to translucent/opaque.

8. Correct Answer : Correct Answer : B — Beam of light

Explanation: A beam is a collection of many light rays traveling together. Beam = group of rays following similar paths.

9. Correct Answer : B — Convergent beam of light

Explanation: When rays meet at a point, they converge. Convergent beam → rays intersect at one point.

10. Correct Answer : B — Reflected ray

Explanation: After striking a surface and returning to same medium, it is called reflected ray. It obeys i = r.

11. Correct Answer : C — Plane mirror

Explanation: Regular reflection occurs on smooth surfaces like plane mirrors. Smoothness ensures single-direction reflection.

12. Correct Answer : A — i = r

Explanation: Law of reflection states angle of incidence equals angle of reflection. Formula: i = r.

13. Correct Answer : D — Collection of light rays

Explanation: Beam of light is defined as a collection of rays. It is not a single path but many rays together.

14. Correct Answer : A — Waves

Explanation: Light is an electromagnetic wave. It follows wave relation c = f λ.

15. Correct Answer : C — Normal

Explanation: Normal is the perpendicular at point of incidence. Angles i and r are measured from the normal line.

16. Correct Answer : C — Point source of light

Explanation: A very tiny source (pinhead size) is treated as a point source. Its size ≈ 0 compared to distance.

17. Correct Answer : B — Mirror

Explanation: Mirror reflects maximum light due to smooth surface. Reflectance is highest compared to paper/cloth/leaf.

18. Correct Answer : C — There is a source of light, an opaque body to obstruct light and a screen.

Explanation: Shadow formation needs source + opaque object + screen. Light blocked → shadow region formed.

19. Correct Answer : B — Transparent

Explanation: Transparent substances transmit most light. Transmission ≫ absorption + reflection.

20. Correct Answer : C — Non-luminous bodies

Explanation: Non-luminous bodies do not emit light; they reflect incident light. Reflected intensity depends on surface.

21. Correct Answer : C — Opaque

Explanation: Opaque bodies do not transmit light. Transmission ≈ 0 through the material.

22. Correct Answer : A — Paper

Explanation: Opaque substances do not allow light to pass. Paper is opaque, while oiled paper/tissue are translucent and plastic may be transparent.

ONE OR MORE THAN CORRECT OPTION TYPE

23. Correct Answer : A, B, D — Plane mirror; Highly polished metals; Distilled water

Explanation: Regular reflection occurs on smooth surfaces where angle of incidence = angle of reflection (i = r). Smooth mirrors, polished metals, and still water act as specular reflectors.

24. Correct Answer : A, B, C — Bulb; Tube light; Burning candle

Explanation: Extended sources have finite emitting area. Bulb, tube light, and candle flame emit from many points, unlike a pin (point source).

25. Correct Answer : B, C, D — Shadows form due to straight-line light; Images show colour; Images are clearer than shadows

Explanation: Shadows are due to rectilinear propagation. Images retain colour and detail; shadows are not always fully black due to penumbra formation.
Invalid (Option A wrong)

26. Correct Answer : B, C — Small image; Inverted image

Explanation: A pinhole camera always forms a real inverted image. In standard school physics, image is diminished (h_i/h_o = v/u < 1); "enlarged" is not accepted here.
Option D not accepted in this level

27. Correct Answer : B, D — Glow worm; Sun

Explanation: Natural luminous sources emit light naturally. Glow worm (bioluminescence) and Sun qualify; candle and torch are artificial.

28. Correct Answer : A, B — Ground glass; Muddy water
Explanation: Translucent substances partially transmit light with scattering. Ground glass and muddy water fit; iron and leather are opaque.

ASSERTION AND REASON TYPE

29. Correct Answer : (A) Both statements I and II are

correct Explanation: The Moon and wood do not emit their own light (Statement I) but become visible by reflecting light from luminous sources (Statement II), fitting the definition of non-luminous bodies. Validity: Valid

30. Correct Answer : (A) Both statements I and II are correct.

Explanation: In optics, a source with negligible dimensions (pinhead size) is a point source, while any source with finite measurable dimensions is an extended source. Validity: Valid

31. Correct Answer : (A) Both assertion and reason are true. The reason is the correct explanation of the assertion

Explanation: A "reflector" is defined as any surface where reflection occurs. Since a plane mirror is designed specifically for this purpose, it fits the definition provided in the Reason. Validity: Valid

32. Correct Answer : (C) Assertion is true but the reason is false. Explanation:

The angle of incidence (i) and angle of reflection (r) are always measured from the **normal** (perpendicular) to the surface, *not* from the mirror surface itself.

33. Correct Answer :

a – S
b – P
c – Q
d – R

Explanation:
Ray of light = single straight ray → S
Parallel beam = multiple parallel rays → P
Divergent beam = rays spread outward from a point → Q
Convergent beam = rays move inward toward a point → R

34. Correct Answer : Answer:

35. Correct Answer : B — Pinhole camera

Explanation: Pinhole camera works on rectilinear propagation — light travels in straight lines to form image.

36. Correct Answer : B — Straight line

Explanation: Rectilinear propagation ⇒ path of light is a straight line.

37. Correct Answer : D —

Explanation: Shadows form because light travels straight and is blocked by objects.

38. Correct Answer : D — All of these

Explanation: U, V, W are vertically symmetric → mirror image looks same (no visible lateral inversion).

39. Correct Answer : B — Z

Explanation: Z has no vertical symmetry → mirror image changes → shows lateral inversion.

40. Correct Answer : D — All of these

Explanation: All given statements are correct examples of lateral inversion.

41. Correct Answer : Direct Answer: 2

Explanation: Laws: (1) ∠i = ∠r, (2) incident ray, reflected ray, and normal lie in same plane.

42. Correct Answer : Direct Answer: 2

PREVIOUS YEAR QUESTIONS

1. Correct Answer : (B)

Explanation: The time shown on clock $\boldsymbol{X}$is 3:25. A plane mirror creates a laterally inverted virtual image, reversing the positions of the hands symmetrically across the vertical 12-to-6 axis. To calculate the mirror time mathematically, subtract the actual time from 11:60. (Formula: $\boldsymbol{T_{mirror} = 11{:}60 - T_{actual}}$)

$$11{:}60 - 3{:}25 = 8{:}35$$

2. Correct Answer : (C) Transparent: (viii), (xi) | Translucent: (ix), (xii) | Opaque: (vii), (x)

Explanation: This can be solved via elimination since metal (i) and books (x) are clearly not transparent, invalidating A, B, and D. Spectacles (viii) and bulb glass (xi) allow light to pass completely (transparent). Tissues (ix) and sunglasses (xii) reduce or scatter light (classified here as translucent). Wooden gates (vii) and books (x) block light entirely (opaque). (Formula: N/A)
Validity: Valid

3. Correct Answer : D

4. Correct Answer : C. The size of image will be the same.

Here is the step-by-step reasoning:

- **Key Property of Plane Mirrors:** A fundamental rule of optics is that a plane (flat) mirror always forms an image that is exactly the **same size** as the object. Mathematically, its magnification (m) is always equal to1.

5. Correct Answer : Answer

6. Correct Answer : B. (ii) and (iv) only.

Here is a breakdown of why each statement is either correct or incorrect:

(i) It is a camera with a single lens. (Incorrect) A pinhole camera is the

simplest type of camera and does not use any lenses at all. It relies purely on a tiny hole (the aperture) to focus light.

(ii) It produces an upside-down image of an object. (Correct) Because light travels in a straight line, the light rays coming from the top of an object pass through the pinhole and hit the bottom of the screen. Conversely, rays from the bottom of the object hit the top of the screen. This results in an inverted (upside-down) image.

(iii) It does not have a screen. (Incorrect) A pinhole camera requires a translucent screen (like tracing paper or frosted glass) on the side opposite the pinhole so that the real image can be projected and viewed.

(iv) It works because light travels in a straight line. (Correct) This principle is known as the rectilinear propagation of light. It is the fundamental reason why the pinhole camera works and why the resulting image is inverted.

(v) It forms virtual and colorless shadow of objects. (Incorrect) A pinhole camera forms a real image (because it is projected onto a screen), not a virtual one. Furthermore, it forms an actual image that retains the colors of the object, not a dark, colorless shadow.

7. Correct Answer : B

5. Electricity and Ciruits

Solutions

WORKSHEET – 1

1. **Correct Answer : B — The filament heats up due to current flow**

   Explanation: Current causes heating effect $\rightarrow H \propto I^2Rt$; hot filament emits light.

2. **Correct Answer : B — Push electric charges through a circuit**

   Explanation: Cell provides potential difference $\rightarrow V = W/Q$to drive charges.

3. **Correct Answer : B — Conductors**

   Explanation: Conductors have low resistance $\rightarrow$ allow current flow $I = V/R$.

4. **Correct Answer : C — An insulator**

   Explanation: Insulators have very high resistance $\rightarrow$ no charge flow.

5. **Correct Answer : B — Safety from electric shock**

   Explanation: Earthing gives low-resistance path $\rightarrow R \downarrow \Rightarrow I$goes to ground.

6. **Correct Answer : B — Cell is exhausted**

   Explanation: No emf/potential difference $\rightarrow$ $I = V/R = 0$.

7. **Correct Answer : A — Both A and R true, R explains A**

   Explanation: Current flows only in closed path $\rightarrow I = V/R$only when circuit complete.

8. **Correct Answer : C — Electric circuit**

   Explanation: Defined as closed conducting path for current flow.

9. **Correct Answer : C — Non-rechargeable**

   Explanation: Primary cells have irreversible reactions $\rightarrow$ cannot restore emf.

10. **Correct Answer : B — Resistance**

    Explanation: Opposition to current $\rightarrow R = V/I$.

WORKSHEET-2

1. **Correct Answer : C — The cell is weak or filament is damaged**

   Explanation: If emf is low or filament is broken $\rightarrow I = \frac{V}{R} \approx 0$, no glow.

2. **Correct Answer : B — Water pump starters**

   Explanation: Push-button switch makes momentary contact $\rightarrow$ completes circuit $I = V/R$.

3. **Correct Answer : C — Chemical energy**

   Explanation: Cell converts chemical energy $\rightarrow$ electrical energy (potential difference V).

4. **Correct Answer : C — Make or break the circuit**

   Explanation: Switch opens/closes path $\rightarrow$ current flows only in closed circuit $I = V/R$.

5. **Correct Answer : B — No closed path is formed between positive and negative terminals**

   Explanation: Both ends at same potential $\rightarrow V = 0 \Rightarrow I = 0$.

6. **Correct Answer : C — Heats up and emits**

   lightExplanation: Heating effect $\rightarrow H \propto I^2Rt$; hot filament glows.

7. **Correct Answer : B — Uses standard symbols to show connections clearly**

   Explanation: Circuit diagrams use symbols to represent components and paths.

8. **Correct Answer : B — Rheostat**

   Explanation: Variable resistance $\rightarrow$ changes Rso current $I = V/R$varies.

9. **Correct Answer : B — Ammeter**

   Explanation: Ammeter measures current → reads I directly.

10. **Correct Answer : B — Resistance box**

    Explanation: Known resistances combined → total $R = R_1 + R_2 + \cdots$.

WORKSHEET-3

1. **Correct Answer : A — Voltage**

   Explanation: In parallel circuits, same potential difference across each branch → V = constant.

2. **Correct Answer : B — Minimum**

   Explanation: Parallel combination → $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots \Rightarrow$ total R decreases.

3. **Correct Answer : C**

4. **Correct Answer : A — Parallel**

   Explanation: Household wiring is parallel → same voltage and independent operation.

5. **Correct Answer : B — Match stick**

   Explanation: Match stick is an insulator → very high $R \Rightarrow I = \frac{V}{R} \approx 0$.

6. **Correct Answer : C**

7. **Correct Answer : A — Electric pole**

   Explanation: Home supply comes from distribution lines on electric poles (mains supply).

8. **Correct Answer : B — The thin filament**

   Explanation: Heating effect → $H \propto I^2Rt$; filament becomes white hot and glows.

9. **Correct Answer : (C) Becomes non-conducting**

   Explanation: Resistance opposes flow, but it does not mean the material stops conducting entirely (which would be an insulator). A resistor still conducts current, just with reduced magnitude.

10. **Correct Answer : B — Decrease**

    Explanation: Ohm's law → $I = \frac{V}{R}$; if R ↑and Vsame ⇒ I ↓.

SINGLE CORRECT OPTION TYPE1:

11. **Correct Answer : C) Electric cell**

    Explanation: An electric cell (battery) converts stored chemical potential energy into electrical potential energy.

12. **Correct Answer : C) Filament**

    Explanation: The filament is the thin, spirally wound tungsten wire that heats up and produces light when current passes through.

13. **Correct Answer : D) Tungsten**

    Explanation: Tungsten is used because it has an extremely high melting point $3422°C$
    allowing it to glow white-hot without melting.

14. **Correct Answer : A**

15. **Correct Answer : B) 2**

    Explanation: A standard electric cell has two terminals: positive (+) and negative (−).

16. **Correct Answer : C) Positive to negative**

    Explanation: By convention, electric current flows from the high potential (positive) terminal to the low potential (negative) terminal outside the source.

17. **Correct Answer : D) Electric circuit**

    Explanation: An electric circuit is the complete, closed path of conductors and components through which electric current can flow.

18. **Correct Answer : C) Electric current**

    Explanation: In the context of this curriculum level, "current" is often the intended answer for what "flows." However, scientifically, current is the rate of flow of charge, not energy itself.
    Validity: Scientifically Imprecise

19. **Correct Answer : B) Earth wire**

Explanation: The earth wire provides a low-resistance path to the ground to prevent electric shock.

20. **Correct Answer : C**

21. **Correct Answer : B) Maximum**

Explanation: In a series circuit, total resistance is $R_{eq} = R_1 + R_2 + \cdots$

22. **Correct Answer : D) Resistance**

Explanation: Resistance is given by $R = \frac{V}{I}$

23. **Correct Answer : B) Insulators**

Explanation: Insulators have tightly bound electrons and very high resistivity.

24. **Correct Answer : D) Plastic**

Explanation: Plastic is an insulator used to jacket wires and appliances to prevent electric shock.

25. **Correct Answer : A) Having excess number of electron**

Explanation: An object becomes negatively charged when it gains excess electrons.

26. **B) Negative to positive**

Explanation: Electrons drift from negative to positive terminal.

27. **Correct Answer : D) Both statements (i) & (ii) are false**

Explanation: Similar charges repel; conductivity depends on free electrons.

28. **Correct Answer : D) Metals are conductors**

Explanation: Metals have free electrons that allow electric current to flow easily.

29. **Correct Answer : B) Series circuit**

Explanation: Components areconnected in a single continuous loop.

30. **Correct Answer : A) Do not glow**

Explanation: If one bulb fails, the circuit becomes open and current stops.

31. **Correct Answer : C) Parallel circuit**

Explanation: Components are connected across the same voltage points.

32. **Correct Answer : D**

ONE OR MORE THAN CORRECT OPTION TYPE

33. **Correct Answer : A) Copper & C) Iron nail**

Explanation: To complete the circuit and allow current to flow (making the bulb glow), the object used must be an electrical conductor. Both Copper and Iron are metals and good conductors.

34. **Correct Answer : B) High melting point & C) Higher resistance**

Explanation: The filament must have high resistance to generate sufficient heat $H = I^2Rt$
and a very high melting point (like Tungsten at $3422°C$) to emit light without melting.

35. **Correct Answer : B) To make & C) To break**

Explanation: The primary function of a switch is to complete ("make") the conducting path to allow current flow or interrupt ("break") the path to stop it.

36. **Correct Answer : B) Voltage is constant & C) Current is divided**

Explanation: In a parallel circuit, the potential difference (voltage) across each component is the same $V_{total} = V_1 = V_2$
while the total current splits among the branches $I_{total} = I_1 + I_2$

ASSERTION AND REASON TYPE

37. **Correct Answer : D) A is false but R is true**

Explanation: The Assertion is false because a bulb glows only when the circuit is closed (complete), not open. An open circuit has a break in the path. The Reason is true: current requires a continuous loop to flow $\rightarrow I = \frac{V}{R}$.

38. Correct Answer : Correct Answer : A) Both A and R are true and R is the correct explanation of A

Explanation: Tungsten has high resistance and a high melting point. As the Reason states, this resistance converts electrical energy into heat → $H = I^2Rt$, raising the temperature until the filament glows white-hot.

39. Correct Answer : C) A is true but R is false

Explanation: The Assertion is true: series bulbs glow dimmer because total resistance increases → $R_{eq} = R_1 + R_2$→ current decreases → $I = \frac{V}{R}$. The Reason is false because it describes a parallel connection, not series.

40. Correct Answer : C) A is true but R is false

Explanation: The Assertion is true: in parallel, each bulb has an independent path and works separately → same branch voltage $V_1 = V_2 = V$. The Reason is false because parallel combination decreases equivalent resistance and increases total current, not the opposite.

MATRIX MATCHING TYPE

41. Correct Answer : Matrix Matching

a) → S) Open circuit
b) → T) Variable resistor
c) → R) Bulb
d) → Q) Closed circuit

42. Correct Answer : a) → T) Rubber

Explanation: Rubber does not allow current to pass easily → high resistance → insulator.
b) → **P) Copper & S) Silver**
Explanation: Copper and silver are best conductors of electricity → very low resistivity → $R = \rho\frac{L}{A}$.
c) → **R) Earthing**
Explanation: Earthing is the process of transferring excess charge to the earth.
d) → **Q) Conductors**
Explanation: Most metals allow current flow → classified as conductors.

COMPREHENSION TYPE

43. Correct Answer : C) Metal cap

Explanation: As stated in the passage, the raised metal cap at the top of a dry cell functions as the positive (+) terminal.

44. Correct Answer : D) Chemical to electrical

Explanation: An electric cell stores energy as chemical potential energy and converts it into electrical energy to drive current through a circuit.

45. Correct Answer : D) Metal disc

Explanation: The passage confirms that the flat metal disc at the bottom of the cell acts as the negative terminal.

46. Correct Answer : C) Electric current

Explanation: Electric current is defined as the rate of flow of electric charge through a conductor per unit time

47. Correct Answer : A) Resistance

Explanation: Resistance is the property of a material that opposes the flow of electrons, often causing the material to heat up (heating effect).

48. Correct Answer : D) Ohmmeter

Explanation: An Ohmmeter is the specific instrument designed to measure electrical resistance (in Ohms).

INTEGER TYPE

49. Correct Answer : 0

Explanation: In a series circuit, if one resistor fails, the circuit becomes open → no current flows → none of the remaining resistors function → $I = 0$.

50. Correct Answer : 0

Explanation: In a parallel circuit, each bulb has an independent path → if one bulb fuses, current still flows in other branches → no other bulbs stop glowing.

PREVIOUS YEAR QUESTIONS

1. Correct Answer : (B)

All the bulbs are not lighting up due to disposable spoon in the circuit which is an non conducting material

2. Correct Answer :

The bulb of circuit (II) would light up the brightest while bulbs of circuit (I) would light up the dimmest as batteries in circuit (II) are in series, bulbs of circuits (III) and (IV) will be equally bright as in circuit (IV) two batteries are connected with similar terminals and effectively it will be equivalent to two batteries connected in series.

3. Correct Answer : (C) :

Bulbs Y and Z are connected with wires making a complete circuit, unlike bulb X. Hence, only Y and Z bulbs will glow when switch is closed.

4. Correct Answer : (D):

When the bulb B_2 blown off, the equivalent resistance of the circuit increases and resultant current passing through bulbs B_1, B_3 and B_5 compared to earlier decreases. As a result, these bulbs get dimmer. While for the Bulb B_4, the current passing through it increases with blown off of B_2, this Bulb get brighter as compared to earlier case.

5. Correct Answer : (C) :

Electrical Energy → T.V. → Walkman

Electrical Energy → Electric bulb → Table Lamp

Electrical Energy → Toaster → Electric Kettle

6. Correct Answer : (B) :

A. All bulbs are in series, it is correct.

B. If Z is removed, the circuit will be incomplete and no bulb will glow.

C. If Z is removed, X and Y will not glow.

D. The given circuit is closed as the path is closed.

7. Correct Answer : (C) :

An electric cell produces electricity from the chemicals stored inside it. Also, combination of electric cells is called a battery.

8. Correct Answer : (B)

When switch S_2 is closed, bulbs B_1 and B_3 will complete the circuit, and thus B_1 and B_3 will glow. B_2 will be disconnected from the circuit and it will not glow.

9. Correct Answer : (C)

in the given circuit, when switch S is opened, lamp 2 and lamp 3 will be switch off.

10. Correct Answer : (B) :

In the shown diagram of torch, 2 represent switch and its function is to make or break the circuit. 3 is a cell and its function is to convert chemical energy into electrical energy. 4 is a spring that makes contact with the switch. 5 is the external body of the torch and protects the device from any external damage. 6 is the torch bulb that glows when the circuit is complete.

6. Magnetism

Solutions

WORKSHEET – 1

1. **Correct Answer: Fe_3O_4**

   Explanation: Natural magnet is magnetite and its chemical formula is Fe_3O_4.

2. **Correct Answer: C) Neodymium**

   Explanation: Neodymium magnets are the strongest permanent magnets among the given options.

3. **Correct Answer: C) Artificial Magnet**

   Explanation: When iron is rubbed with magnetite, it becomes magnetized and is called an artificial magnet.

4. **Correct Answer: A) Magnetic length = 5/6 × Geometric length**

   Explanation: Magnetic length is slightly shorter than geometric length and is approximately 5/6 of it.

5. **Correct Answer: B) Magnetic pole**

   Explanation: Magnetic poles are located slightly inside the ends of a bar magnet where magnetic force is strongest.

6. **Correct Answer: C) Magnetic axis**

   Explanation: The imaginary line joining the north and south poles is called the magnetic axis.

7. **Correct Answer: A) Equatorial line**

   Explanation: The equatorial line passes through the midpoint of the magnet and is perpendicular to the magnetic axis.

8. **Correct Answer: C) South pole**

   Explanation: The end of a magnet that points toward the geographic south is called the south pole.

9. **Correct Answer: A) 15 cm**

   Explanation: Magnetic length = (5/6) × 18 cm = 15 cm.

10. **Correct Answer: C) Magnetic length**

    Explanation: The distance between the two magnetic poles of a magnet is called its magnetic length.

11. **Correct C) Reverse**

    Explanation:

    In the electrical method (like using a current-carrying coil or solenoid), the direction of the magnetic field depends on the direction of current.

    When the current direction is reversed, the magnetic field direction also reverses.

12. **Correct Answer: B) Two**

    In the double touch method, two magnets are used simultaneously to magnetize a piece of material.

WORKSHEET-2

1. **Correct Answer: C) Diamagnetic material**

   Diamagnetic materials are repelled by a magnet. Examples include bismuth and mercury.

2. **Correct Answer: A) Paramagnetic**

   Paramagnetic materials are weakly attracted by magnets. They do not retain magnetism after the field is removed.

3. **Correct Correct Answer: B) Non-magnetic material**

A non-magnetic material is neither attracted by a magnet nor can it be magnetised (like wood, plastic, glass).

4. **Correct Answer: B) Magnetic field**

The region around a magnet where its magnetic force can be felt is called the magnetic field.

5. **Correct Answer: A) Ferromagnetic**

Ferromagnetic materials (like iron, cobalt, nickel) are strongly attracted by magnets and can be permanently magnetised.

6. **Correct Answer: A) Magnetic compass**

A magnetic compass is used to find directions. It works using Earth's magnetic field.

7. **Correct Answer: A) Aluminium, nickel and cobalt**

Alnico is made from Aluminium, Nickel, and Cobalt (that's how the name comes

8. **Correct Answer: A) Hydrogen, mercury**

Hydrogen and mercury are examples of diamagnetic materials.
Iron and nickel are ferromagnetic, and platinum is paramagnetic.

9. **Correct Answer: B) North Pole to South Pole**

Outside a magnet, magnetic field lines go from the North Pole to the South Pole.

10. **Answer: C) Is neither attracted by a**

magnet nor can be converted into an artificial magnet
That's the definition of a non-magnetic substance.

WORKSHEET-3

1. **Correct Answer: D) Magnetic sticker**

Electromagnets need electric current to work. TV, electric fan, and electric bell use electromagnets. A magnetic sticker is usually a permanent magnet, not an electromagnet.

2. **Correct Answer: D) Generator**

A generator uses magnets to produce electricity through electromagnetic induction. Microscope, periscope, and telescope mainly use lenses and mirrors.

3. **Correct Answer: B) Demagnetisation**

When a magnet loses its magnetic properties, the process is called demagnetisation.

4. **Correct Answer: A) Fans, motors, and mixers**

explanation: These appliances use **electric motors**, and motors work using **electromagnets** to produce motion.

5. **Correct Answer: B) Attract**

Unlike poles (North–South) attract each other. Like poles repel.

6. **Correct Answer: D) All of these**

Heating, hammering, and dropping disturb the alignment of magnetic domains, causing demagnetisation.

7. **Correct Answer: B) North–South direction**

Again, due to Earth's magnetic field, a freely suspended magnet aligns North–South.

8. **Correct Answer: C) Repulsion**

Repulsion is the sure test of magnetism because attraction can occur with magnetic and non-magnetic materials, but repulsion only happens between two magnets.

9. **Correct Answer: B) Points toward Earth's**

magnetic north
The north-seeking pole points toward the Earth's magnetic north region.

10. **Correct Answer: C) Converting magnetic material into a magnet**
Magnetisation is the process of turning magnetic material (like iron) into a magnet.

SINGLE CORRECT OPTION TYPE

1. **Correct Answer: D) All of these**
Bar magnet, horseshoe magnet, and magnetic needle are all man-made (artificial) magnets.
2. **Correct Answer: A) Y must be a magnet**
Repulsion is the sure test of magnetism. Only a magnet can repel another magnet.
3. **Correct Answer: B) Magnetite**
Magnetite (lodestone) is a natural magnet and an ore of iron.
4. **Correct Answer: B) Artificial magnets**
Magnets made by human methods are called artificial magnets.
5. **Correct Answer: A) Magnetic materials**
Materials attracted by magnets are called magnetic materials.
6. **Correct Answer: C) Both A & B**
Manganese and platinum are paramagnetic materials.
7. **Correct Answer: A) Iron**
Iron is ferromagnetic. Hydrogen, manganese, and aluminium are not ferromagnetic.
8. **Correct Answer: C) Less than**
Magnetic length is slightly less than geometric length.
9. **Correct Answer: D) Aluminium**
Nickel, dysprosium, and cobalt are ferromagnetic; aluminium is not.
10. **Correct Answer: A) Mica**
Mica is non-magnetic. (Aluminium is weakly magnetic.)
11. **Correct Answer: B) Soft iron**
Temporary magnets are made of soft iron because it magnetises easily and loses magnetism quickly.
12. **Correct Answer: D) Steel**
Permanent magnets are made of steel due to high magnetic retention.
13. **Correct Answer: B) Temporary magnet**
Temporary magnets lose magnetism easily — poor retention.
14. **Correct Answer: C) Pole**
Each end of a bar magnet is called a pole.
15. **Correct Answer: C) Magnetic meridian**
The vertical plane passing through magnetic North–South is the magnetic meridian.
16. **Correct Answer: B) North pole**
The end pointing toward geographic north is called the north pole.
17. **Correct Answer: B) Ferrite is a strong magnetic material**
Air is diamagnetic (not paramagnetic), so option C is incorrect. Hence not "All of the above."
18. **Correct Answer: B) They can push or pull some objects**
without touching
Magnetic substances exert force without contact.
19. **Correct Answer: C) Near the ends**
Magnetic strength is maximum at poles.
20. **Correct Answer: B) Magnetisation**
Converting magnetic material into a magnet is magnetisation.
21. **Correct Answer: B) Slightly inside**
Magnetic poles are located slightly inside the ends.

22. **Correct Answer: C) Pins stick at the ends of the magnet**
Maximum magnetic force is at poles.

23. **Correct Answer: D) Iron**
Iron is ferromagnetic, not paramagnetic.

24. **Correct Answer: B) Closed curves**
Magnetic field lines always form closed loops.

25. **Correct Answer: B) Artificial**
Artificial magnets can be very strong and made in desired shapes.

26. **Correct Answer: (Depends on figure)**
Since the figure isn't shown, direction depends on pole positions. Without the diagram, exact rotation cannot be determined.

27. **Correct Answer: C) 6 cm**
Magnetic length = 5 cm
Geometric length ≈ Magnetic length + small correction (~20%)
So geometric length ≈ 6 cm (approx.)

28. **Correct Answer: D) B and C**
Strength is maximum at both North and South poles.

29. **Correct Answer: B) Huge and weak magnet**
Earth acts like a giant but weak magnet.

30. **Correct Answer: C) Weakest, strongest**
Magnet is weakest at the middle and strongest at the ends.

31. **Correct Answer: B) 12 cm**
Magnetic length ≈ 0.8 × Geometric length
So Geometric length = 10 / 0.8 = 12.5 ≈ 12 cm

ONE OR MORE THAN CORRECT OPTION TYPE

32. **Correct Answer: A, B, C**
Heating, hammering, and dropping disturb magnetic domains and cause demagnetisation. So all three are correct.

33. **Correct Answer: A, C, D**
Cobalt, iron, and steel are magnetic materials and can be used to make magnets.
Ebonite is non-magnetic, so it cannot be magnetised.

34. **Correct Answer: B, C, D**
A comb, book, and wood are non-magnetic materials, so they do not get attracted to a magnet.
An iron nail does get attracted.

35. **Correct Answer: A, B**
Ferrite magnets are commonly used in radios and transistors because they are strong and resist demagnetisation.
They are not typically used in street lights or vehicles for that purpose.

ASSERTION AND REASON TYPE

36. **Correct Answer: C) I and II are not correct**
Explanation:
- Statement I is incorrect because a magnet has both attraction and repulsion properties. Like poles repel and unlike poles attract.
- Statement II is incorrect because iron can become a magnet and is strongly attracted by magnets.
- Statement III is also incorrect because magnetic field lines emerge from the North pole and enter the South pole, not the other way around.

37. **Correct Answer: C) Both I and II are correct**
Explanation:
- Statement I is correct. Repulsion is considered the sure test of magnetism, because attraction can occur with magnetic and non-magnetic materials, but repulsion only occurs between two magnets.
- Statement II is correct. Only similar poles (N–N or S–S) repel each other.

38. Correct Answer: D) I → Correct, II → Incorrect

Explanation:

I → Correct

II → Incorrect

39. Correct Answer: B) Both A and R are true but R is not the correct explanation of A.

- Explanation:
- Assertion (A) is true: Ferromagnetic materials are strongly attracted by magnets.
- Reason (R) is also true: Iron and nickel are examples of ferromagnetic materials.
- However, R only gives examples; it does not explain *why* they are strongly attracted. So R is not the correct explanation.

MATRIX MATCHING TYPE

40. Correct Answer:

a) Poles facing each other → Horse-shoe magnet (S)

b) Closed curve from north to south outside magnet → Magnetic line of force (P)

c) Area of magnetic influence → Magnetic field (T)

d) Small needle for tracing lines → Tracing compass (R)

e) Device for finding direction → Magnetic compass (Q)

41. Correct Answer:

a) Strongest magnet → Q) Neodymium
b) Diamagnetic materials → P) Gold, bismuth
c) Ferromagnetic material → S) Nickel, cobalt, dysprosium
d) Paramagnetic material → R) Manganese, aluminium, air

COMPREHENSION TYPE

42. Correct Answer: B) Repels

Explanation: Like poles repel each other. North pole near North pole → repulsion.

43. Correct Answer: C) A force of repulsion

Explanation: When two North poles are brought close, they repel due to like poles repelling.

44. Correct Answer: A) A force of attraction

Explanation: Unlike poles attract each other. South pole near North pole → attraction.

45. Correct Answer: B) North-South

Explanation: A freely suspended magnet always aligns itself along the North–South direction because Earth behaves like a giant magnet.

46. Correct Answer: A) Geometrical North

Explanation: The north-seeking end of a magnet points toward the geographical (geometrical) North.

47. Correct Answer: B) Geometrical South

Explanation: The south pole of the magnet points toward the geographical (geometrical) South.

INTEGER TYPE

48. Correct Answer: 2

Explanation: Every magnet has two poles — one North and one South.

49. Correct Answer: 5 / 6

Explanation:
Magnetic length = (Geometric length × 5/6)
Magnetic length is slightly less than geometric length (approximately 5/6 of it).

50. Correct Answer: 1

Explanation: Only one magnetic meridian passes through a freely suspended magnet — the vertical plane along which it aligns in the North-South direction.

PREVIOUS YEAR QUESTIONS

1. Correct Answer : is B)

The field is strongest just outside the poles, and R is the closest point to a pole.

2. Correct Answer : is B):

Both needles point in the same direction (→ →)

3. Correct Answer : The incorrect statement is: C.

Magnetic force is a pulling force only.

4. Correct Answer : B

- X → N
- Y → S
- Z → N

5. Correct Answer : C. I and III only

- I: A, C, E, G → all same (N)
- III: C (N) and H (S) → attract

6. Correct Answer : is: C

- Original magnet: N on left, S on right
- So each broken piece (I, II, III, IV) will also have:
  N on left and S on right

7. Beyond Earth

Solutions

WORKSHEET – 1

1. **Correct Answer : is D) Constellations.**
Constellations are groups of stars that form recognizable patterns in the sky. People in ancient times used these patterns to identify different regions of the night sky.

2. **Correct Answer : is C) Blue star.**
The color of a star shows its temperature. Blue stars are the hottest, while red and orange stars are cooler in comparison.

3. **Correct Answer : is A) Earth's atmosphere bends the starlight.**
As starlight passes through different layers of Earth's atmosphere, it keeps bending. This makes stars appear to flicker or twinkle.

4. **Correct Answer : is C) Sun.**
The Sun is a star because it produces its own light and heat. The other options are groups or patterns of stars, not a single star.

5. **Correct Answer : is C) Ursa Major.**
Ursa Major is a well-known constellation made of several stars forming a pattern. Betelgeuse and Aldebaran are individual stars.

6. **Correct Answer : is A) To find the North direction.**
The Pole Star (Polaris) stays almost fixed in the northern sky. Travelers and sailors used it to locate the North direction for navigation.

7. **Correct Answer : is C)**
Excess artificial lights, along with smoke and dust, hide the stars.
Bright city lights create light pollution, which reduces visibility of faint stars. Smoke and dust also scatter light and make the sky hazy.

8. **Correct Answer : is B) The sky becomes completely black, making faint stars easier to see.**
On a moonless night there is less light in the sky. This darkness helps us see dim stars and other celestial objects clearly.

9. **Correct Answer : is A)**
To allow their eyes to adjust to the darkness so faint stars become visible. Our eyes need about 20–30 minutes to adapt to darkness. After adjustment, we can see more faint stars.

10. **Correct Answer : is B) Sirius.**
Orion's Belt forms a straight line in the sky. Extending this line toward the east points to Sirius, the brightest star in the night sky.

WORKSHEET-2

1. **Correct Answer : is C) Planets shine with steady light.**
Planets do not produce their own light; they reflect sunlight and usually appear steady. Stars twinkle because their light passes through Earth's atmosphere and gets disturbed.

2. **Correct Answer : is C) Gases.**
The outer planets like Jupiter and Saturn are called gas giants because they are made mostly of hydrogen and helium gases rather than solid rock.

3. **Correct Answer : is B) Rotation.**

Rotation is the movement in which Earth spins on its own axis. This spinning causes day and night on Earth.

4. **Correct Answer : is A) Revolution.**

Revolution is the movement of Earth around the Sun along its orbit. One complete revolution takes about 365 days, forming one year.

5. **Correct Answer : is B) Between Mars and Jupiter.**

The Asteroid Belt is a region filled with many rocky objects. It lies mainly between the orbits of Mars and Jupiter.

6. **Correct Answer : is B) Vikram and Pragyan.**

In Chandrayaan-3, Vikram is the lander that touched the Moon's surface, and Pragyan is the rover that moved around to study the surface.

7. **Correct Answer : is D) Dust, rocks, ice, and frozen gases.**

Comets are called "dirty snowballs" because they contain a mixture of ice, frozen gases, dust, and small rocky particles.

8. **Correct Answer : is B) Its ice and frozen gases start changing into gas.**

When a comet comes closer to the Sun, heat causes its ice to vaporize into gas, forming a glowing coma and often a tail.

9. **Correct Answer : is C) 1986 and 2061.**

Halley's Comet was last visible from Earth in 1986. It is a periodic comet and is expected to return again around 2061.

10. **Correct Answer : is C) Periodic comets.**

Comets that return repeatedly after fixed time intervals while moving in a regular orbit around the Sun are called periodic comets.

WORKSHEET-3

1. **Correct Answer : is A) Stars, planets, gas, and dust.**

A galaxy is a huge system made up of many objects, not just stars. It also includes planets, clouds of gas, and dust, all held together by gravity.

2. **Correct Answer : is B) It is a small part of the Milky Way Galaxy.**

Our Solar System is located within the Milky Way, not outside it. It is just one tiny part among billions of stars in this galaxy.

3. **Correct Answer : is C) Ākāśha Gangā.**

The glowing milky band seen in the night sky is our home galaxy. In many Indian traditions, the Milky Way is also called Ākāśha Gangā.

4. **Correct Answer : is B) Milky Way Galaxy.**

Earth and the entire Solar System exist inside the Milky Way. It is the galaxy that contains our Sun and nearby stars.

5. **Correct Answer : is D) Spiral shape with long arms.**

The Milky Way is shaped like a spiral with curved arms extending outward. These arms contain stars, gas, and dust.

6. **Correct Answer : is D) We are inside the galaxy.**

We see the Milky Way as a bright band because we are viewing it from within. Looking across its dense star regions creates the milky appearance.

7. **Correct Answer : is B) Thick regions of stars, gas, and dust.**

New stars form in dense clouds called nebulae. These regions contain gas and dust that collapse under gravity to create stars.

8. **Correct Answer : is C) The Universe is expanding.**

If galaxies are moving away from each other, it shows that space itself is stretching. This observation supports the idea of an expanding universe.

9. **Correct Answer : is B) Distances in space are extremely large.**

Space distances are so vast that kilometers become inconveniently large numbers. Light-years provide a more practical way to express these huge distances.

10. **Correct Answer : is A) Solar System → Milky Way → Universe.**

The Solar System is a small part of the Milky Way galaxy. The Milky Way itself is just one galaxy within the much larger Universe.

SINGLE CORRECT OPTION TYPE

1. **Correct Answer : IS C**

Stars appear tiny because they are at extremely large distances from Earth. Even though many are bigger than Earth (and even the Sun), their vast distance makes them look like small points.

2. **Correct Answer : IS B**

Ancient people used constellations as guides for navigation, especially during sea travel and desert journeys. Their fixed patterns helped determine direction at night.

3. **Correct Answer : IS A**

Betelgeuse is traditionally associated with the Indian name **Ārdrā**. The other options mismatch well-known identifications like Polaris (Dhruva) and Orion (a constellation, not Kṛittikā).

4. **Correct Answer : IS D**

A village field on a moonless night has minimal light pollution and no bright moonlight. Dark skies allow the faint light from many more stars to be visible.

5. **Correct Answer : IS D**

The Big Dipper has two “pointer stars” that form an imaginary line leading toward the Pole Star. This method has been traditionally used to locate Polaris in the sky.

6. **Correct Answer : IS B**

Sky-mapping apps use your time and location to show accurate positions of stars and constellations. They act as digital guides to help identify what is visible in the sky.

7. **Correct Answer : IS C**

The Milky Way looks like a faint band because we are inside the galaxy viewing the combined light of billions of distant stars. Their blended light forms the milky glow.

8. **Correct Answer : IS B**

If the Moon had an atmosphere, wind and weather could slowly erode and smooth crater edges over time. Currently, craters remain sharp because there is no air or weather.

9. **Correct Answer : IS C**

Chandrayaan-3 successfully landed near the Lunar South Pole region. This mission demonstrated India’s capability for a soft landing on the Moon.

10. **Correct Answer: (A) Ice and gases refreeze and stop escaping**

A comet’s tail forms when heat from the Sun causes its ice to turn into gas and dust. When the comet moves away, it becomes colder, the gases refreeze, and the tail fades or disappears.

11. **Correct Answer: (A) Earth is a satellite of the Sun**

A satellite is any object that revolves around a bigger body. Earth revolves around the Sun, so in this definition Earth acts as a (natural) satellite of the Sun. The other options are incorrect because the Sun does not orbit Earth, the Moon orbits Earth (not the Sun), and Jupiter does not orbit Mars.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

12. **Correct Answer: is A, B and D.**

A planet is described as a massive, nearly round object that travels in an orbit around the Sun. It also rotates on its axis as part of its motion. Producing its own light is a property of stars, not planets.

13. **Correct Answer: is A, B and D.**

The inner planets are Mercury, Venus, Earth, and Mars, which lie closer to the Sun. Jupiter is an outer planet, so it is not included among the inner planets.

14. **Correct Answer: is A and B.**

A natural satellite is any object that orbits a larger body in space. The Moon orbiting Earth and planets orbiting the Sun are natural satellites, while GPS and communication satellites are man-made.

15. **Correct Answer: is A, B and D.**

The Big Dipper helps find the Pole Star by using the two pointer stars. An imaginary line is drawn through them and extended several times their distance to locate the Pole Star.

16. **Correct Answer: is A, B and D.**

Planets shine with steady light and usually do not twinkle like stars. They can often be seen with the naked eye, but they do not produce their own light—they reflect sunlight.

17. **Correct Answer: is A, B, C and D.**

Outer planets are much larger, mostly made of gases, and generally very cold due to their distance from the Sun. They are located beyond Mars in the outer region of the Solar System.

ASSERTION AND REASON TYPE

18. **Correct Answer: is A.**

Natural satellites are bodies that orbit planets, which makes the Assertion true. The Moon is a natural satellite that orbits Earth, and this example correctly explains what a natural satellite is.

19. **Correct Answer: is A.**

Most asteroids are found in the region between Mars and Jupiter, so the Assertion is true. This region is called the Asteroid Belt, which directly explains where most asteroids are located.

20. **Correct Answer: is A.**

Venus is the hottest planet, so the Assertion is true. Its thick atmosphere traps heat through a strong greenhouse effect, which explains why it becomes extremely hot.

21. **Correct Answer: is A.**

Jupiter, Saturn, Uranus, and Neptune are called giant planets, making the Assertion true. They are very large and mostly made of gases (or icy gases), which explains why they are grouped as giant planets.

MATRIX MATCHING TYPE

22. **Correct Answer: A–S, B–Q, C–R, D–P**
23. **Correct Answer: A–R, B–Q, C–S, D–P**

COMPREHENSION TYPE

24. Correct Answer: B) Stars, planets, gas, and dust

Explanation: A galaxy is a huge system made up of many stars along with their planets, clouds of gas, and dust, all held together by gravity.

25. Correct Answer: B) Milky Way Galaxy

Explanation: The Solar System, including Earth and the Sun, is located inside the Milky Way Galaxy.

26. Correct Answer: C) Everything that exists

Explanation: The Universe is the largest concept—it includes all galaxies, stars, planets, space, matter, and energy; in short, everything that exists.

27. Correct Answer: C) They are trillions of kilometers away

Explanation: Stars look tiny because they are extremely far from Earth, even though many of them are actually much larger than our planet.

28. Correct Answer: B) Constellations

Explanation: When groups of stars form shapes or patterns in the sky, they are called constellations, which people used for stories and navigation.

29. Correct Answer: C) Polaris (Pole Star)

Explanation: The Pole Star (Polaris) stays almost fixed in the northern sky and helps travelers identify the north direction.

INTEGER TYPE

30. Correct Answer: 24 hours

Earth rotates on its axis once approximately every 24 hours. This rotation causes **day and night**—when your part of Earth faces the Sun it is day, and when it turns away it becomes night.

31. Correct Answer: 8 planets

Our Solar System has eight planets: **Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune**. Pluto was once considered a planet but is now classified as a **dwarf planet**, so it is not included in the main count.

32. Correct Answer:

5th position

PREVIOUS YEAR QUESTIONS

1. Correct Answer : is C)

Venus rotates backward on its axis as compared to most of the other planets. This means on Venus, the Sun rises in the West and sets in the East. Venus is the second nearest planet to the Sun.

2. Correct Answer : is D) The closest celestial object to the Earth is Moon.

The galaxy to which our solar system belongs is the **Milky Way**.

Recognisable patterns of stars in the sky are called **constellations**.

The phase when the Moon is totally invisible is called the **New Moon**.

So,
P → (iii), Q → (iv), R → (ii), S → (i)

3. Correct Answer : is B)

Venus is called as the morning star as well as evening star, depending on its position with respect to the sun.